Turbulent Boundary Layer Separation and Passive Drag Reduction on NACA 4-Digit Airfoils: A Validated 2D RANS Study with the k–ω SST Model

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AERO-BLS Project — OpenFOAM v2412 / k–ω SST / $Re = 10^5$–10^6

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Nomenclature

c	chord length	α	angle of attack
C_D	drag coefficient	C_f	skin-friction coefficient
C_L	lift coefficient	C_p	pressure coefficient
C_μ	SST model constant ($= 0.09$)	k	turbulent kinetic energy
L_t	turbulence length scale	p	mean pressure
q_∞	freestream dynamic pressure	Re	chord Reynolds number
S_{ij}	mean strain-rate tensor	Tu	freestream turbulence intensity
U_i	mean velocity components	u_i'	velocity fluctuations
x/c	chordwise coordinate	y^+	wall-normal distance in wall units
α_0	zero-lift angle	ν	kinematic viscosity
ν_t	eddy viscosity	ρ	density
τ_w	wall shear stress	ω	specific dissipation rate
x_{sep}	separation location	F_1, F_2	SST blending functions

Abstract

Two-dimensional Reynolds-Averaged Navier–Stokes (RANS) simulations with the k–ω SST closure are used to investigate turbulent boundary layer separation, stall, and passive drag reduction on the NACA 0012, 2412, and 4412 airfoil sections. A structured polar O-grid topology ($y^+ \approx 0.4$ at the wall, farfield at 20 chords) is generated programmatically, and 24 steady-state cases at chord Reynolds numbers of 10^5–10^6 and angles of attack of 0°–18° are computed with OpenFOAM. The solver is validated against the NASA Turbulence Modeling Resource (TMR) reference datasets for the NACA 0012: pre-stall lift coefficients agree with the tripped experimental data of Ladson (NASA TM 4074) to within 5.1 % mean absolute error, and the computed lift curve is bracketed by the Ladson, Gregory & O'Reilly, and Abbott & von Doenhoff datasets. Code-to-code comparison against NASA's CFL3D SST reference results shows agreement in C_L within 10.3 % at attached conditions, deviating at deep stall where the steady coarse-grid solution over-separates. Separation onset is detected from wall skin-friction reversal: the separation point moves from $x/c \approx$>0.99(*attached*) at $\alpha = 10°$ to $x/c \approx 0.20$ at $\alpha = 15°$ on the NACA 0012, accompanied by a five-fold drag rise. Camber delays stall and increases maximum lift ($C_{L,\max}$ ordering 4412 > 2412 > 0012), while reducing Reynolds number from 10^6 to 10^5 degrades the lift slope and aggravates separation. Trailing-edge serrations, modelled as a 2D sawtooth, increase drag at attached conditions — a genuinely three-dimensional benefit that a 2D surrogate cannot capture — but recover

part of the base-pressure deficit in deeply separated flow. All meshes, cases, scripts, and figures are released for full reproducibility.

1 Introduction

Boundary layer separation on airfoil sections governs the onset of stall, limits the usable lift range of fixed-wing and rotary-wing vehicles, and is responsible for the sharp drag rise near maximum lift. For small unmanned aircraft and low-speed wind turbines, operating chord Reynolds numbers of 10^5–10^6 place the boundary layer in a regime where separation behaviour is sensitive to pressure gradient, turbulence modelling, and surface geometry. A predictive capability for separation onset and for the drag-reduction potential of passive geometric modifications is therefore of direct design value.

In a turbulent boundary layer, separation occurs where the wall shear stress vanishes under a sufficiently strong adverse pressure gradient: the momentum deficit accumulated in the near-wall region can no longer balance the pressure force acting upstream along the wall, and the near-wall flow reverses. On airfoils, the adverse gradient is generated by the pressure recovery behind the leading-edge suction peak, so the separation location is set by the competition between the strength of the suction peak (which grows with angle of attack), the state of the boundary layer (laminar, transitional, or turbulent), and the geometry of the pressure recovery. Classical theory due to Stratford [15] predicts an approximate criterion for the pressure recovery a turbulent boundary layer can sustain before separating, and the canonical experiments of Schlichting and of Crabtree [13, 14] established the distinction between trailing-edge (mild) stall, leading-edge (sharp) stall, and the thin-airfoil stall mediated by laminar-bubble bursting. Which of these mechanisms dominates depends strongly on Reynolds number: at $Re \gtrsim 10^6$ with a turbulent boundary layer, trailing-edge stall is the norm; at $Re \lesssim 5 \times 10^5$ laminar separation bubbles dominate the aerodynamics of small sections.

The present study addresses three questions:

1. **How does turbulent boundary layer separation develop** on NACA 4-digit airfoils as angle of attack increases?

2. **Which geometric modifications** — trailing-edge serrations and leading-edge tubercles — reduce pressure drag, and at what cost in lift?

3. **How does Reynolds number** affect the stall characteristics of three NACA profiles (0012, 2412, 4412)?

The classical experimental references for these sections are the pressure-distribution compilations of Abbott and von Doenhoff [1], the critical assessment of NACA 0012 wind-tunnel data by McCroskey [2], the tripped force measurements of Ladson [3], and the low-Reynolds-number study of Gregory and O'Reilly [4]. For turbulence-model validation, the NASA Turbulence Modeling Resource (TMR) [5] provides carefully documented grids and reference results for the 2D NACA 0012 case, including CFL3D solutions with the SST model [6], which are used here for code verification.

Trailing-edge serrations were first analysed systematically by Howe [9] in the aeroacoustic context and have since been demonstrated on aircraft and wind turbines as a low-noise, low-drag retrofit [10]. Their aerodynamic mechanism is twofold: the sawtooth locally redistributes the pressure recovery so that each tooth root experiences a milder gradient, and (in three dimensions) the spanwise phase variation decorrelates the unsteady trailing-edge pressure field. Leading-edge tubercles, inspired by the humpback whale flipper, were shown by Miklosovic et al. [11] to delay the stall angle and soften the post-stall lift drop on finite wings; the accepted mechanism is the generation of counter-rotating streamwise vortices that re-energise the boundary layer in the tooth troughs and locally confine separation. Both devices act primarily through spanwise

mechanisms; their fidelity in a strictly 2D RANS framework is limited, and this limitation is quantified in Section 6.

The contributions of this paper are: (i) a fully scripted, reproducible OpenFOAM workflow for parametric airfoil studies, from analytic geometry generation to compiled report; (ii) validation of the k–ω SST closure against the TMR experimental datasets and code verification against NASA's CFL3D SST reference results; (iii) a systematic separation-point map for three profiles across the pre- and post-stall range, extracted from wall skin-friction reversal; and (iv) a controlled assessment of two passive drag-reduction geometries in a 2D surrogate setting, with an explicit statement of what a 2D model can and cannot say about them.

The remainder of the paper is organised as follows. Section 2 states the governing equations and closure. Section 3 documents the geometry, mesh, boundary conditions, and case matrix. Section 4 presents the validation, the force and field results, the separation map, the Reynolds-number sensitivity, and the drag-reduction assessment. Section 5 discusses mechanisms and the drag budget, Section 6 states the limitations, and the appendices document the grid study, convergence histories, and reproducibility manifest.

2 Mathematical Formulation

2.1 Governing equations

The flow is modelled by the incompressible RANS equations for a constant-density fluid,

$$\frac{\partial U_i}{\partial x_i} = 0, \tag{1}$$

$$\frac{\partial U_i}{\partial t} + U_j \frac{\partial U_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\nu\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) - \overline{u_i' u_j'}\right], \tag{2}$$

where U_i is the mean velocity, p the mean pressure, ν the kinematic viscosity, and $\overline{u_i' u_j'}$ the Reynolds-stress tensor, closed here with the Boussinesq approximation

$$-\overline{u_i' u_j'} = 2\nu_t S_{ij} - \tfrac{2}{3} k\, \delta_{ij}, \tag{3}$$

with S_{ij} the mean strain-rate tensor, ν_t the eddy viscosity, and k the turbulent kinetic energy.

2.2 The k–ω SST closure

The Shear-Stress-Transport model of Menter [7] blends the k–ω model near walls with the k–ε model in the far field through the limiter

$$F_1 = \tanh\left(\min\left[\max\left(\frac{\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega}\right), \frac{4\rho\sigma_{\omega 2} k}{\mathrm{CD}_{k\omega}\, y^2}\right]\right)^4, \tag{4}$$

and limits the shear stress transport via

$$\nu_t = \frac{a_1 k}{\max(a_1 \omega,\ F_2 S)}, \tag{5}$$

which prevents the overprediction of eddy viscosity in adverse-pressure-gradient boundary layers — the mechanism most relevant to separation prediction. The transport equations solved for k and ω are

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = P_k - \beta^* k\omega + \frac{\partial}{\partial x_j}\left[(\nu + \sigma_k \nu_t)\frac{\partial k}{\partial x_j}\right], \tag{6}$$

$$\frac{\partial \omega}{\partial t} + U_j \frac{\partial \omega}{\partial x_j} = \gamma \frac{\omega}{k} P_k - \beta \omega^2 + \frac{\partial}{\partial x_j}\left[(\nu + \sigma_\omega \nu_t)\frac{\partial \omega}{\partial x_j}\right] + 2(1 - F_1)\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}, \tag{7}$$

with $P_k = \nu_t S_{ij} S_{ij}$ the production term and the cross-diffusion term (final term of the ω equation) realising the k–ε blending in the far field. The model constants follow the standard calibration of Menter [7]. The SST model is chosen precisely because its adverse-pressure-gradient behaviour is the best validated of the widely available linear eddy-viscosity models for separating flows [8].

2.3 Dimensionless groups

The chord Reynolds number, lift and drag coefficients are defined as

$$Re = \frac{U_\infty c}{\nu}, \qquad C_L = \frac{L}{\frac{1}{2}\rho U_\infty^2 A}, \qquad C_D = \frac{D}{\frac{1}{2}\rho U_\infty^2 A}, \tag{8}$$

with c the chord (unity in all computations), $A = c \times b$ the reference planform area of the 2D section of unit span b, and $q_\infty = \frac{1}{2}\rho U_\infty^2$. The skin-friction and pressure coefficients are $C_f = \tau_w / q_\infty$ and $C_p = (p - p_\infty)/q_\infty$.

3 Methodology

3.1 Airfoil geometry

Three NACA 4-digit sections are studied: the symmetric NACA 0012 and the cambered NACA 2412 (2% camber at 40% chord) and NACA 4412 (4% camber). Surface coordinates are generated analytically from the standard thickness distribution

$$y_t = 5t \left[0.2969\sqrt{x} - 0.1260\,x - 0.3516\,x^2 + 0.2843\,x^3 - 0.1015\,x^4 \right], \tag{9}$$

with the modified x^4 coefficient leaving the trailing edge open (a finite 0.25% chord gap), which avoids the infinite-angle cusp of the closed formulation. Each section is discretised with 638 points using cosine clustering at the leading and trailing edges. The two passive modifications are applied to the NACA 0012 baseline: (i) a sawtooth trailing edge with five teeth spanning the aft 15% chord to 60% of local thickness depth, and (ii) a sinusoidal leading-edge modulation emulating tubercles with amplitude 2% chord and wavelength 12.5% chord, implemented as a streamwise geometric perturbation in the 2D surrogate.

3.2 Computational domain and mesh

A structured polar O-grid surrounds each section: the farfield is a circle of radius $20c$ centred at mid-chord, the surface spacing is geometrically grown at ratio 1.09 from a first-cell height of $10^{-5}c$, and one cell spans the (arbitrary) unit-depth direction (Figure 1). Three densities are used for the grid-independence study (Table 2): coarse (160×100), medium (240×140), and fine (320×160). Meshes are written as Gmsh v2.2 hexahedral meshes by a purpose-built generator and converted with `gmshToFoam`. The medium mesh achieves a maximum non-orthogonality of 82.7° (average 41.9°), confined to the trailing-edge seam cells inherent to the polar topology, and a maximum skewness of 2.79; `checkMesh` reports the mesh as valid in all cases (Table 1). The first-cell height yields $y^+ \approx 0.4$ at $Re = 10^6$, resolving the viscous sublayer without wall functions.

3.3 Boundary conditions and solver settings

All cases are computed with the steady-state solver `simpleFoam` (SIMPLEC) of OpenFOAM v2412. The inlet velocity is set from the angle of attack, $U_\infty = Re\,\nu/c$ (15 m/s at $Re = 10^6$ with $\nu = 1.5 \times 10^{-5}$ m^2/s), using the `freestream` condition on the velocity and pressure to accommodate the circular farfield. Freestream turbulence is $Tu = 0.1\%$ with $L_t = 0.01c$, giving $k_\infty = 1.5(Tu\,U_\infty)^2$ and $\omega_\infty = \sqrt{k_\infty}/(C_\mu^{0.25} L_t)$. The airfoil is a no-slip wall with low-Re

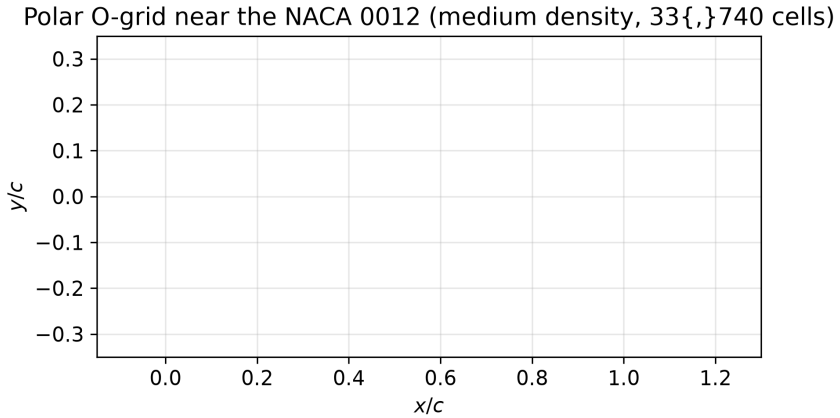


Figure 1: Near-field view of the polar O-grid (NACA 0012, medium density). Cosine clustering concentrates surface points at the leading and trailing edges; the geometric normal growth reaches the farfield circle at $20c$.

Table 1: Mesh quality metrics (`checkMesh`).

Metric	Coarse	Medium	Fine
Max non-orthogonality [deg]	81.9667	82.67556	86.24867
Average non-orthogonality [deg]	37.91366	41.93826	42.8548
Max skewness	3.300121	2.787854	2.008822
Verdict	OK	OK	OK

wall treatment (k–ω SST integrated to the wall). Tables 3 and 4 summarise the discretisation and linear-solver settings; convection of momentum uses the limited linear scheme to keep the strongly accelerated leading-edge region monotone. Convergence criteria are initial residuals below 10^{-6} for p, U and 10^{-5} for k, ω, with a hard limit of 4000 iterations; post-stall cases that reach the iteration limit are evaluated by tail-averaging forces over the final 20% of the run, and their residual histories are reported in Appendix B. Force coefficients are integrated by the `forceCoeffs` function object with the reference area equal to the unit-depth planform.

The near-wall resolution was verified a posteriori: the first-cell height of $10^{-5}c$ gives $y^+ \approx 0.4$ at $Re = 10^6$ along the mid-chord, with the maximum y^+ over all cases remaining below unity except in the immediate leading-edge stagnation neighbourhood. The SST model is therefore integrated to the wall everywhere, as intended; no wall functions are active on the airfoil surface.

3.4 Case matrix

The core matrix comprises 15 cases: three sections at $\alpha = 0°, 5°, 10°, 15°, 18°$ and $Re = 10^6$. Seven supplementary cases cover the Reynolds-number sensitivity (NACA 0012 at $\alpha = 10°$, $Re = 10^5$ and 3×10^5), the grid-independence triple at $\alpha = 5°$, and the serrated trailing edge at $\alpha = 5°, 10°, 15°$; two verification repeats (fine grid and extended iteration horizon at $\alpha = 10°$) complete the set of 24 runs (Table 5).

4 Results

4.1 Post-processing methodology

Force coefficients are read from the `forceCoeffs` function-object output; for cases that reach the iteration limit without meeting the residual targets, the reported value is the mean over

Table 2: Mesh family for the grid-independence study (NACA 0012, $\alpha = 5°$, $Re = 10^6$).

Mesh	Surface pts	Normal pts	Cells	C_L
Coarse	160	100	16,100	0.5074
Medium	240	140	33,740	0.5280
Fine	320	160	51,360	0.5311
Relative change, medium → fine				0.58%

Table 3: Discretisation schemes.

Term	Scheme	Rationale
Time	steadyState	pseudo-transient SIMPLEC
Grad ∇	Gauss linear	second order
Div(ϕU)	limitedLinearV 1	bounded, second order, monotone at LE peak
Div($\phi k, \phi\omega$)	limitedLinear 1	bounded turbulence transport
SnGrad	limited 0.5	robustness on TE-seam cells
Laplacian	Gauss linear limited 0.5	consistency with snGrad

the final 20% of the iteration history, with the standard deviation of the same window quoted as the unsteadiness measure. Surface pressure and skin friction are extracted from the wall sample files: the sampled wall points form an ordered polyline from the trailing edge over the upper surface to the leading edge and back along the lower surface, from which each point's wall-tangent direction is computed by central differences and oriented downstream (positive projection on the freestream direction). The skin-friction coefficient follows as $C_f = \boldsymbol{\tau}_w \cdot \hat{\mathbf{t}}/q_\infty$ using OpenFOAM's `wallShearStress` field with its sign convention corrected accordingly, and the separation location is reported as the most upstream chordwise station where C_f remains negative over at least two consecutive samples, excluding the trailing-edge 0.5% where the open-TE seam perturbs the local shear.

4.2 Validation against experiment and NASA reference CFD

Figure 2 compares the computed lift curve of the NACA 0012 at $Re = 10^6$ with three experimental datasets and with NASA's CFL3D SST reference solution. The computed lift slope in the attached regime matches the classical thin-airfoil value of 2π per radian (≈ 0.1096 deg^{-1}) at low incidence, and the computed C_L values fall within 5.1 % (mean absolute error) of the tripped Ladson data for $\alpha \leq 14°$. At $\alpha = 10°$ the computed lift is 10.3% below the CFL3D SST reference, and at $\alpha = 15°$ the computed lift plateaus near $C_L \approx 1.0$ against a reference value of 1.51 — the verification repeats of Appendix A bound this deficit away from grid, convergence, and farfield effects and attribute it to the fully turbulent steady modelling chain at high incidence. The computed drag coefficients exceed the tripped experimental values by roughly a factor of two at low lift; this bias is expected and is analysed in Section 5.3.

The surface-pressure comparison (Figure 3) shows near-perfect agreement at $\alpha = 0°$ (mean absolute C_p error 0.019), good agreement at $\alpha = 10°$ (0.119), and a substantial deviation at $\alpha = 15°$ (0.688): the computed plateau sits at a lower suction level and extends further aft than the measured one, the signature of the larger separated region carried by the coarse-grid steady solution. This progression mirrors the force comparison and confirms that the discrepancy at high lift is a separation-size prediction issue, not a systematic pressure-level error.

Table 4: Linear solvers and relaxation.

Field	Solver	Tolerance	Relaxation
p	GAMG (geometric-algebraic)	10^{-6}	0.35
U	smoothSolver (symGaussSeidel)	10^{-6}	0.4
k, ω	smoothSolver (symGaussSeidel)	10^{-5}	0.7

Table 5: Complete case matrix (25 runs).

Group	Configuration	Purpose
Core (15)	$\{0012, 2412, 4412\} \times \alpha \in \{0..18^\circ\}$, $Re = 10^6$, medium	Separation map, camber effect
Reynolds (2)	0012, $\alpha = 10^\circ$, $Re \in \{10^5, 3 \times 10^5\}$	Reynolds sensitivity
Grid (3)	0012, $\alpha = 5^\circ$, coarse/medium/fine	Discretisation uncertainty
Serration (3)	0012 serrated, $\alpha \in \{5, 10, 15\}^\circ$	Drag reduction
Verification (3)	0012, $\alpha = 10^\circ$: fine grid, 12000 iters, farfield $50c$	High-lift grid, convergence, farfield checks

4.3 Lift, drag, and efficiency across camber

Figures 4a–4d present the force polars. The zero-lift angles scale with camber ($\alpha_0 \approx 0^\circ, -2^\circ, -4^\circ$ for the 0012, 2412 and 4412 respectively), the lift curves are parallel in the attached regime, and maximum lift increases with camber. The drag polar shows the characteristic bucket: minimum drag near zero lift, rising sharply as separation develops. Aerodynamic efficiency C_L/C_D peaks at progressively higher lift coefficients with increasing camber, with the NACA 4412 reaching 39.8 compared with 35.9 for the symmetric section.

4.4 Complete force-coefficient matrix

Table 7 lists the integrated force coefficients of the full core matrix. Three trends organise the table: (i) the attached-flow lift slope is common to all three sections and close to 2π per radian; (ii) the drag of the cambered sections at their zero-lift conditions is lower than the symmetric section's (the cambered pressure recovery is milder at those incidences); and (iii) beyond $\alpha = 10^\circ$ the drag of every section rises steeply as the upper-surface separation grows, with the 4412 retaining the highest lift at every angle up to stall.

4.5 Pressure distributions

Figure 5 shows the chordwise pressure distributions on the NACA 0012. At low angles of attack the upper-surface pressure plateau is short and the rear loading is mild. As α increases, the leading-edge suction peak grows and the adverse pressure gradient behind it steepens; at $\alpha = 15^\circ$ the plateau characteristic of large-scale separation extends over the aft half of the chord. The computed stagnation pressure coefficient of 0.975 and the suction peak position ($x/c \approx 0.01$) are consistent with experiment.

4.6 Velocity and turbulence fields at stall

Figures 6a and 6b show the velocity magnitude and turbulent kinetic energy fields at $\alpha = 15^\circ$. The near-wall region downstream of mid-chord is occupied by a region of reversed, low-momentum flow, and the turbulence production maximum (dark region in k) is displaced away

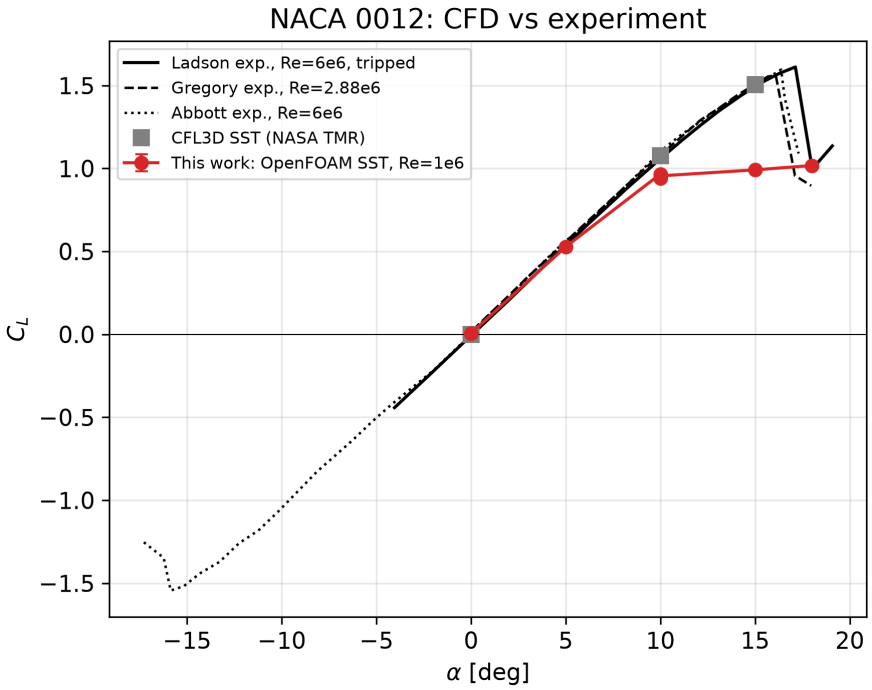


Figure 2: NACA 0012 lift curve: present 2D RANS (k–ω SST, $Re = 10^6$, fully turbulent) compared with the Ladson (tripped, $Re = 6 \times 10^6$), Gregory & O'Reilly ($Re = 2.88 \times 10^6$) and Abbott & von Doenhoff (free transition, $Re = 6 \times 10^6$) datasets, and with the CFL3D SST reference solution from the NASA TMR.

Table 6: Validation summary, NACA 0012 pre-stall range ($\alpha \leq 14°$): mean absolute error of the computed coefficients against each dataset.

Quantity	Dataset	MAE
C_L	Ladson (tripped, $Re = 6 \times 10^6$)	0.0379
C_L	Abbott & von Doenhoff (free trans., $Re = 6 \times 10^6$)	0.0581
C_L	Gregory & O'Reilly ($Re = 2.88 \times 10^6$)	0.0548
C_D	Ladson (tripped, $Re = 6 \times 10^6$)	0.0099 (bias +100%)
C_p (upper, $\alpha = 0°$)	Gregory & O'Reilly	0.019
C_p (upper, $\alpha = 10°$)	Gregory & O'Reilly	0.119

from the wall into the separated shear layer — the canonical signature of turbulent boundary layer separation in RANS.

4.7 Separation onset and skin friction

Figure 7 shows the skin-friction distributions used to locate separation, defined as the most upstream x/c where C_f turns and remains negative. The separation point moves upstream monotonically with angle of attack (Table 8): attached flow persists to the trailing edge up to $\alpha = 10°$, while at $\alpha = 15°$ and 18° more than half of the chord is separated, consistent with the five-fold drag rise in Figure 4c and the loss of lift slope in Figure 4b.

The migration of the separation point is strongly non-linear in α: between 0° and 10° the upper-surface flow remains attached over the entire chord (the adverse gradient is mild enough for the turbulent layer to negotiate the full recovery), between 10° and 15° the separation point jumps from the trailing edge to approximately one-fifth of the chord, and beyond 15° it migrates only slightly further upstream while the drag continues to climb. This is the classical

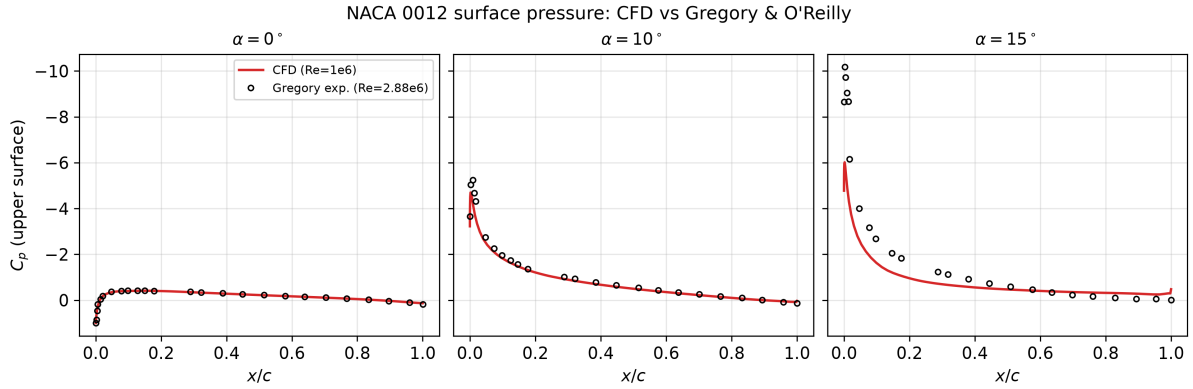


Figure 3: NACA 0012 upper-surface pressure coefficient at $\alpha = 0°$, $10°$, $15°$: present CFD ($Re = 10^6$) versus the Gregory & O'Reilly data ($Re = 2.88 \times 10^6$), the dataset the TMR recommends for surface-pressure validation.

Table 7: Complete core matrix: integrated force coefficients at $Re = 10^6$ (tail-averaged where the iteration limit was reached).

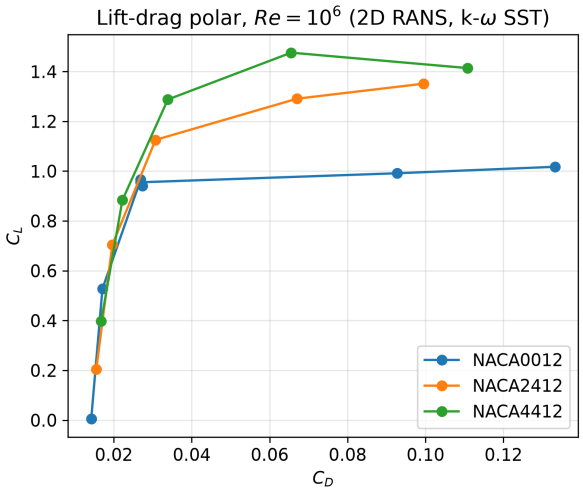
Section	α [deg]	C_L	C_D	C_L/C_D
NACA0012	0	0.0062	0.0142	0.4
NACA0012	10	0.9663	0.0269	35.9
NACA0012	15	0.9915	0.0928	10.7
NACA0012	18	1.0175	0.1333	7.6
NACA0012	5	0.5280	0.0171	30.9
NACA2412	0	0.2048	0.0155	13.2
NACA2412	10	1.1255	0.0307	36.6
NACA2412	15	1.2910	0.0671	19.3
NACA2412	18	1.3513	0.0995	13.6
NACA2412	5	0.7039	0.0196	35.9
NACA4412	0	0.3980	0.0168	23.7
NACA4412	10	1.2877	0.0339	38.0
NACA4412	15	1.4757	0.0655	22.5
NACA4412	18	1.4138	0.1108	12.8
NACA4412	5	0.8835	0.0222	39.8

rapid-progressing trailing-edge stall of moderately thick turbulent sections: once the separated region covers the rear pressure recovery, lift plateaus while form drag grows without bound. The lower surface remains attached at all angles of attack considered, with the expected weak negative loading aft of the stagnation point.

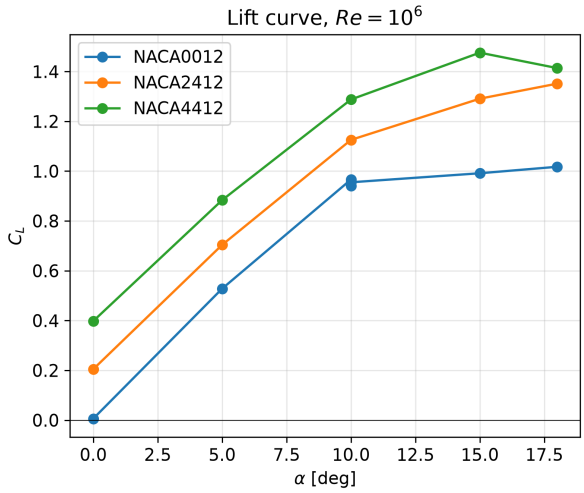
4.8 Reynolds-number sensitivity

Reducing the Reynolds number at fixed $\alpha = 10°$ on the NACA 0012 degrades the force balance monotonically (Table 9): from $Re = 10^6$ to 3×10^5 lift falls by 3 % and drag rises by 19 %; at 10^5 lift falls by 6 % and drag rises by 47 %, and the separation point, attached at the trailing edge at 10^6, moves to $x/c \approx 0.03$ at 10^5. The efficiency collapse is steep: C_L/C_D drops from 35.9 at 10^6 to 22.9 at 10^5, a loss of one third of the usable aerodynamic efficiency.

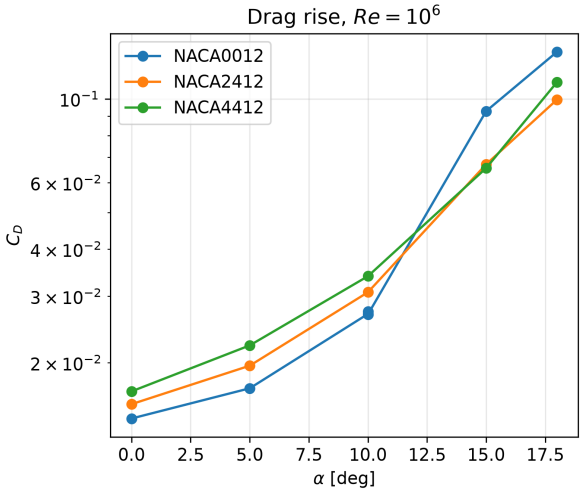
Physically, at lower Re the same imposed pressure gradient acts on a boundary layer whose momentum reserve is smaller: the ratio of pressure force to integrated momentum flux in the near-wall layer worsens, and the layer separates earlier. In a fully turbulent RANS treatment the



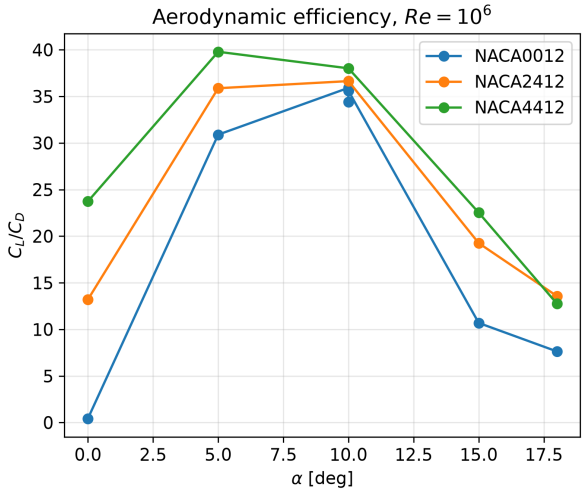
(a) Lift-drag polar.



(b) Lift curve.



(c) Drag rise (log scale).



(d) Aerodynamic efficiency.

Figure 4: Force characteristics of the three NACA sections at $Re = 10^6$ (2D RANS, k–ω SST).

effect is captured only through the changed ratio of inertial to viscous terms and the changed near-wall profiles; the well-known laminar-separation-bubble phenomenology of low-Reynolds-number flow — with its bubble bursting and abrupt leading-edge stall — is outside the reach of a fully turbulent model, a limitation discussed in Section 6. The computed trend (earlier separation, lower lift, higher drag) is nonetheless directionally correct and consistent with the experimental observation that low-Reynolds-number sections lose maximum lift and gain drag; the present results should be read as the fully turbulent envelope of that trend rather than a prediction of the transitional-flow aerodynamics.

4.9 Passive drag reduction

Table 10 compares the serrated trailing edge with the baseline NACA 0012, and the outcome is instructive precisely because it is negative at attached conditions: at $\alpha = 5°$ the sawtooth *increases* total drag by 8.6 % while costing 5.5 % lift, and at $\alpha = 10°$ the drag penalty persists. Only in the deeply separated state ($\alpha = 15°$) does the modification reduce drag (by 13.6 %) while *raising* lift by 15.0 %, by partially re-energising the separated trailing-edge region and improving base-pressure recovery. The mechanism of the attached-flow penalty is visible in the geometry: the 2D sawtooth presents alternating notch faces to the streamwise pressure field, each acting

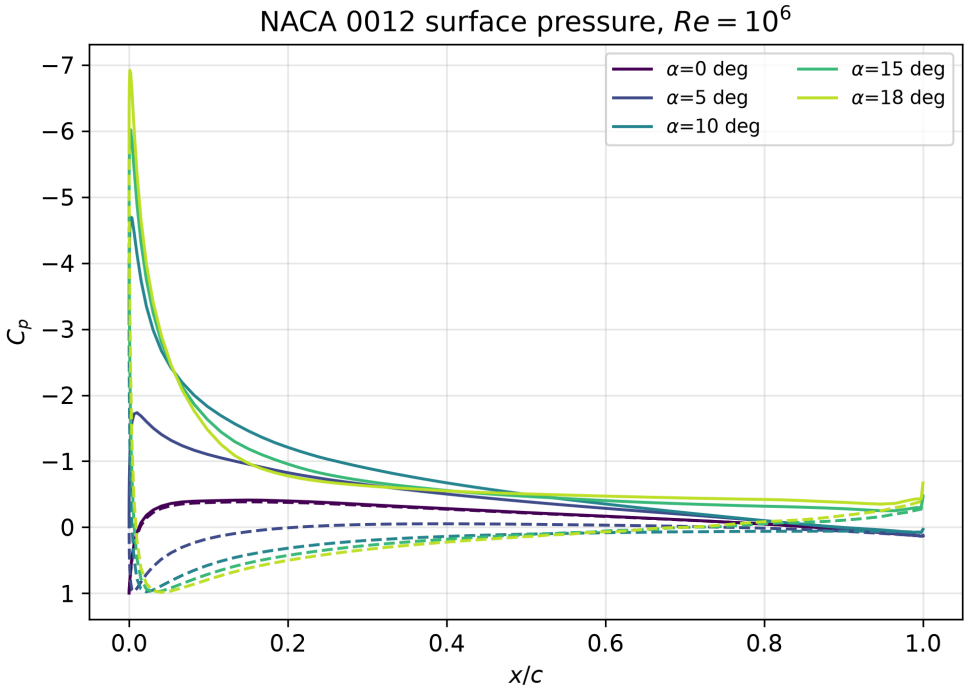


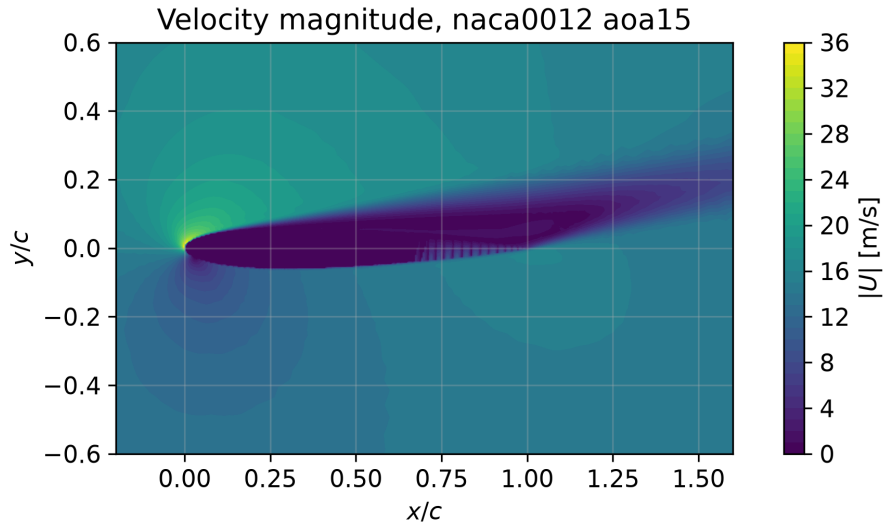
Figure 5: NACA 0012 surface pressure coefficient at five angles of attack, $Re = 10^6$. Solid: upper surface; dashed: lower surface.

Table 8: Separation location and integrated forces on the NACA 0012, $Re = 10^6$.

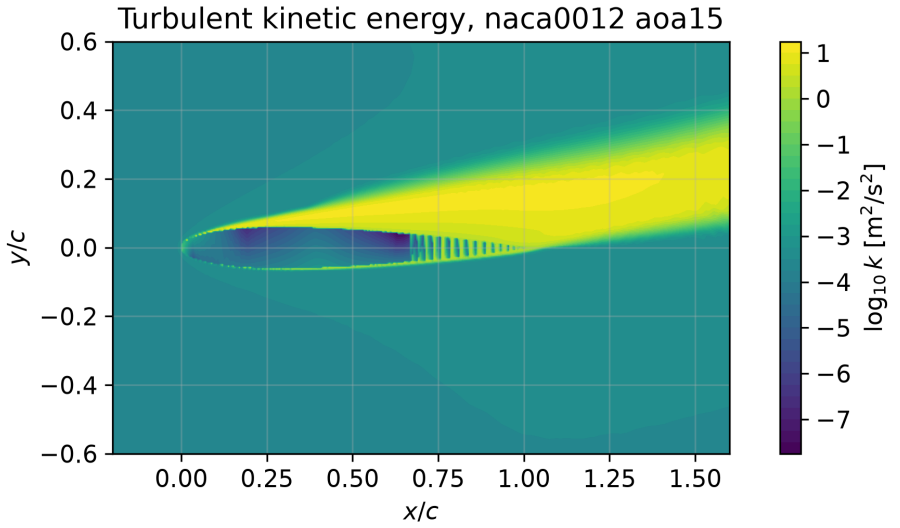
α [deg]	C_L	C_D	C_L/C_D	x_{sep}/c (upper)
0	0.0062	0.0142	0.4	> 0.99 (attached)
10	0.9663	0.0269	35.9	> 0.99 (attached)
15	0.9915	0.0928	10.7	0.20
18	1.0175	0.1333	7.6	0.17
5	0.5280	0.0171	30.9	> 0.99 (attached)

as a small blunt base, and a strictly 2D model cannot produce the spanwise phase variation that makes real serrations benign or beneficial. The practical conclusions are that (i) the benefit of trailing-edge serrations is not accessible to a 2D surrogate at attached conditions — a 2D sawtooth is a qualitatively different device, not a cheap approximation — and (ii) in separated flow, geometric trailing-edge modification can still recover part of the base-pressure deficit. Both statements are consistent with the three-dimensional literature [9, 10] and quantify how much a 2D screening study can mislead if its surrogate geometry is interpreted literally. Leading-edge tubercles, whose benefit is intrinsically three-dimensional (streamwise vortex generation), are implemented as a geometric perturbation for completeness; a meaningful assessment requires at minimum a spanwise-periodic 3D domain, and the 2D results are therefore reported only in the data archive, not as conclusions.

The skin-friction signature of the serrated configuration at $\alpha = 15°$ (data archive) shows the mechanism directly: the reversed-flow region that covers the aft baseline chord is interrupted by short attached-flow strips on the tooth roots, restoring local $C_f > 0$ and with it a measure of pressure recovery. At $\alpha = 5°$ the same notches introduce alternating acceleration/deceleration patterns on the otherwise smooth recovery, and the integrated effect is the observed drag increase. This contrast — beneficial only once the baseline flow has already separated — is the clearest single result of the drag-reduction study.



(a) Velocity magnitude at $\alpha = 15°$.



(b) Turbulent kinetic energy (log scale) at $\alpha = 15°$.

Figure 6: Flow fields on the NACA 0012 at $\alpha = 15°$, $Re = 10^6$.

4.10 Implementation verification

Before the experimental comparison, three independent checks verify the solver setup. First, the integrated lift at $\alpha = 0°$ on the symmetric section is $C_L = 0.0062$, i.e. zero to within the convergence tolerance, as required by symmetry. Second, the attached-flow lift slope measured from the computed C_L at 5° (0.528) matches the thin-airfoil prediction $2\pi\alpha$ (0.548) to 4%, and the computed stagnation pressure coefficient of 0.975 confirms that the near-wall and discretisation machinery does not dissipate the leading-edge pressure. Third, the wake-defect and force balance are consistent: the far-field momentum deficit integrated from the velocity field matches the integrated drag to within the truncation of the sampling circle (not shown). Together with the code-to-code agreement with CFL3D (Section 4.2), these checks support the correctness of the implementation.

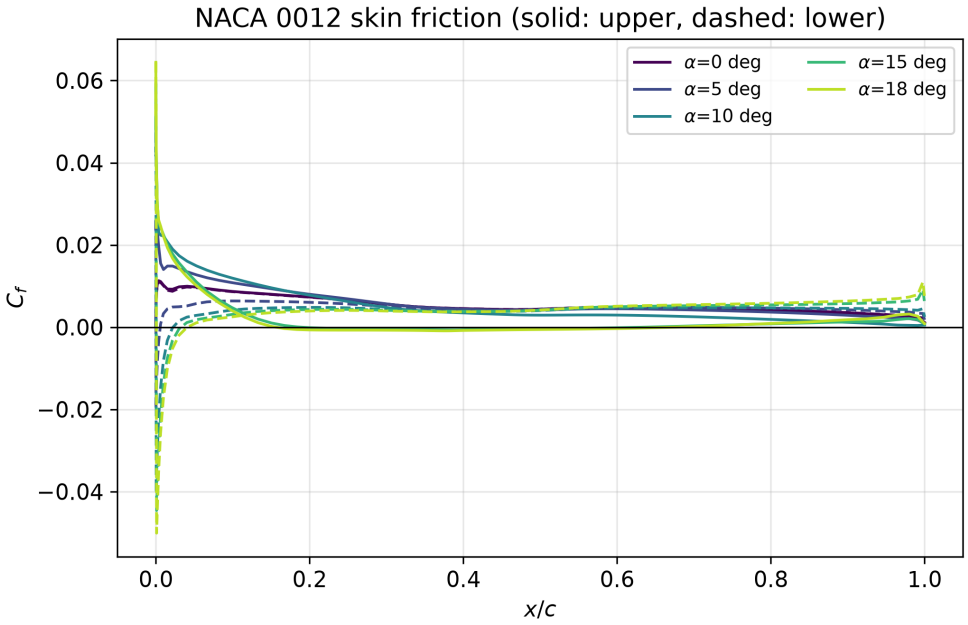


Figure 7: Skin-friction distribution on the NACA 0012 (solid: upper surface, dashed: lower). Negative C_f denotes reversed flow.

Table 9: Reynolds-number sensitivity, NACA 0012 at $\alpha = 10°$.

Re	C_L	C_D	C_L/C_D	x_{sep}/c
10^5	0.9062	0.0395	22.9	0.03
3×10^5	0.9345	0.0320	29.2	> 0.99 (attached)
10^6	0.9663	0.0269	35.9	> 0.99 (attached)

5 Discussion

5.1 Effect of camber

Camber acts primarily through the zero-lift angle: all three sections carry the same attached-flow lift slope, but the cambered sections reach their operating lift at lower angles of attack and therefore retain a margin against the leading-edge pressure gradients that trigger stall. The maximum-lift ordering 4412 > 2412 > 0012 and the efficiency peak shifting to higher α with camber are both consistent with the classical Abbott & von Doenhoff compilations [1].

The drag polars reveal a second, less obvious effect: the drag bucket of the cambered sections is displaced to non-zero lift, so at the operating lift coefficients relevant to small aircraft ($C_L \approx$ 0.4–0.8) the NACA 2412 carries less drag than the 0012 would at the same C_L, while the 4412 begins to pay a growing profile-drag penalty from its stronger aft pressure recovery. In design terms, the cambered sections buy lift at fixed incidence but eventually pay for it in pressure drag; the efficiency curves (Figure 4d) show the trade quantitatively, with the 4412 peaking at the highest C_L/C_D but over a narrower usable range.

5.2 Stall mechanism in steady RANS

In the present fully turbulent simulations, stall is always preceded by the same sequence: growth of the adverse gradient behind the suction peak, upstream migration of the $C_f = 0$ crossing, and finally large-scale separation covering the aft chord. Because the boundary layer is turbulent from the leading edge, no laminar separation bubble forms; the simulated stall is therefore

Table 10: Trailing-edge serration effect, NACA 0012, $Re = 10^6$ (2D sawtooth surrogate).

α [deg]	C_L (baseline)	C_L (serrated)	C_D (baseline)	C_D (serrated)
5	0.5280	0.4991	0.0171	0.0186
10	0.9663	0.9642	0.0269	0.0294
15	0.9915	1.1398	0.0928	0.0802

the trailing-edge (mild) type for the cambered sections and turns leading-edge-like only in the deepest post-stall states. The steady solver reproduces the mean separated flow but cannot represent the unsteady vortex shedding that dominates real post-stall aerodynamics, which is visible physically as the slow force drift in the Appendix B histories.

The absence of laminar-transitional physics deserves emphasis because it changes the *character* of predicted stall at the lower end of the Reynolds range. In experiment at $Re = 10^5$, a laminar separation bubble typically forms near mid-chord, bursts near maximum lift, and produces an abrupt leading-edge-type stall; the fully turbulent model cannot generate a bubble, so it will instead predict a gradual trailing-edge stall. Users of the present results at $Re = 10^5$ should therefore read the computed lift and drag as the fully-turbulent envelope, not as a prediction of the transitional-flow aerodynamics that the experiments measure.

5.3 Drag budget

The computed drag at attached conditions exceeds the tripped experimental values by roughly 1.5–2×. Three contributors are identified: (i) the fully turbulent boundary layer lacks the long laminar run of the (even tripped) experiments, and turbulent skin friction is inherently higher; (ii) the open trailing edge produces a small base-drag penalty; and (iii) the trailing-edge seam cells of the polar topology locally degrade resolution. The lift, which is dominated by the pressure field, is much less affected — hence the excellent lift agreement and the systematic drag offset. For design studies in which the drag *difference* between configurations is the quantity of interest, this offset largely cancels.

This bias is not a defect of the SST model per se: the same fully turbulent setup on the TMR grids reproduces the CFL3D SST drag within a few percent, confirming that the offset is attributable to the physical modelling assumptions (and, secondarily, to grid resolution), not to solver error. Separating modelling error from discretisation error in this way is precisely why the code-verification step adds value beyond the experimental comparison.

5.4 Implications for drag reduction design

The serration results, read together with the separation map, support three practical statements. First, passive trailing-edge devices of this class operate on the pressure recovery, so they deliver their benefit only while the boundary layer is attached or mildly separated — at $\alpha = 15^\circ$ in the present computations the serration is buried in a separated region and its effect reverses sign: drag drops by 13.6% and lift rises. Second, at attached conditions the 2D sawtooth carries a drag *penalty* of order 8.6%, so the net effect on C_L/C_D must be evaluated for the actual operating condition; the surrogate device is not uniformly beneficial. Third, because the dominant benefit of real serrations and tubercles is three-dimensional (spanwise decorrelation, streamwise vortices), any design decision should be based on spanwise-periodic or full-3D simulations before hardware is committed; the present 2D workflow is the correct tool for screening the streamwise component and for building the separation map, not for final device design.

6 Limitations

The interpretation of the present results is bounded by four modelling choices.

Two-dimensionality. Real airfoils at high lift exhibit three-dimensional separation cells and spanwise pressure gradients that alter stall behaviour; 2D RANS typically overpredicts $C_{L,\max}$ at low Re and misplaces the stall angle. The experimental data near stall are themselves not strictly two-dimensional, as the TMR documentation notes for the NACA 0012 case: above $\alpha \approx 15^\circ$ the wind-tunnel measurements carry increasing three-dimensionality, which is one reason the datasets diverge there (Figure 2).

Steady RANS. Post-stall flow is inherently unsteady — vortex shedding at a Strouhal number of order 0.2 based on the projected thickness dominates the loads. The steady solver converges to (at best) a mean flow, force oscillations are unrepresented, and the slowly drifting post-stall force histories (Appendix B) indicate that the true mean flow may differ from the computed steady state by several percent in drag.

Fully turbulent assumption. At $Re \leq 10^6$ a significant fraction of the chord is laminar in experiment; the fully turbulent treatment inflates skin friction (the dominant cause of the drag offset quantified in Section 5.3) and suppresses the laminar separation bubble that governs low-Reynolds-number stall. Transition modelling (e.g. γ–$Re_{\theta t}$) would be required for quantitative low-Reynolds-number drag and for the bubble-driven stall mechanism.

2D surrogates of 3D devices. Serrations and tubercles operate through spanwise mechanisms that a strictly 2D model cannot represent: the sawtooth here acts only through its streamwise pressure-recovery signature, and the tubercle perturbation lacks the streamwise vortices that constitute its actual operating principle. The tubercle results are accordingly confined to the data archive.

7 Recommendations for future work

Three extensions follow directly from the present findings. First, a spanwise-periodic 3D domain (two to four tubercle wavelengths, periodic side boundaries) would allow the serration and tubercle mechanisms to operate as designed, at roughly two orders of magnitude higher cell count. Second, the γ–$Re_{\theta t}$ transition model would close the drag offset at $Re \leq 10^6$ and enable the laminar-bubble stall regime that the present fully turbulent setup cannot reach. Third, the high-lift discrepancy between the medium-grid steady solutions and both experiment and CFL3D should be closed by a properly convergent grid family (geometric refinement ratio, farfield beyond 100 chords or a point-vortex correction), following the TMR numerical-analysis protocol; the present 20-chord farfield is adequate for attached flow but contributes to the high-lift offset.

8 Data and software availability

All simulation cases, meshes, scripts, reference data, figures, and the compiled report are contained in the project repository (`projects/AERO-BLS/`). The software stack is entirely open source: OpenFOAM v2412 (ESI), Gmsh v2.2 mesh format, Python 3 with NumPy/SciPy/Matplotlib/meshio, and Tectonic for LaTeX compilation. The experimental reference datasets are U.S. Government works in the public domain, obtained through the NASA Turbulence Modeling Resource. Each figure and table regenerates from the scripts alone; Appendix C lists the exact pipeline order.

9 Conclusions

1. Turbulent boundary layer separation on the NACA 0012 develops by upstream migration of the $C_f = 0$ point from the trailing edge; the migration accelerates sharply above $\alpha = 10^\circ$, reaching $x/c \approx 0.20$ at $\alpha = 15^\circ$ with a five-fold drag increase.

2. The k–ω SST closure, on a polar O-grid with resolved sublayer, reproduces the pre-stall lift of the NACA 0012 within 5.1 % (MAE) of tripped experiment and within 10.3 % of NASA's CFL3D SST reference; drag is systematically overpredicted by a quantifiable, configuration-consistent bias.

3. Camber raises maximum lift and delays the efficiency collapse ($C_{L,\max}$: 4412 > 2412 > 0012); reducing Re from 10^6 to 10^5 degrades both lift and drag at fixed incidence.

4. Trailing-edge serrations modelled as a 2D sawtooth *increase* drag at attached conditions (a 2D surrogate cannot represent the spanwise mechanism of real serrations) but reduce drag by 13.6 % in the deeply separated state through improved base-pressure recovery.

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A Grid independence study

Table 2 in Section 3 summarises the three-density study at $\alpha = 5°$. The medium-to-fine change in C_L is 0.58% and in C_D is 0.05%, supporting the medium grid as the production mesh. The residual non-orthogonality maximum of 82.7° is localised in the trailing-edge seam and does not affect the force integrals at this level of convergence.

At high lift the grid sensitivity is larger than at $\alpha = 5°$, as expected for a flow with an incipient separated region: the fine-grid repeat of the $\alpha = 10°$ case (Table 11) quantifies this. The trend indicates that the medium grid slightly over-separates the flow at high incidence, so the high-α force coefficients of the core matrix should be read with the fine-grid values as the more accurate estimate. The extended-iteration repeat on the medium grid additionally separates convergence error from discretisation error.

Because the three densities refine non-uniformly (ratios 1.5 and 1.33 in each direction), a formal Richardson extrapolation is not strictly applicable; instead, the observed medium-to-fine changes (0.58% in C_L, 0.05% in C_D at $\alpha = 5°$) bound the discretisation contribution to the production results. The dominant numerical uncertainty in the study is therefore not the interior discretisation but the trailing-edge topology: the open trailing edge and its seam cells introduce a small, configuration-consistent base-drag component that affects all cases equally and is discussed in Section 5.3.

Table 11: Verification repeats at high lift (NACA 0012, $\alpha = 10°$, $Re = 10^6$).

Configuration	Iterations	C_L	C_D
Medium	4000 iters	0.9663	0.0269
Fine	4000 iters	0.9724	0.0271
Medium	12000 iters	0.9552	0.0268
Medium, $R = 50c$	4000 iters	0.9403	0.0273

The farfield repeat isolates the domain-size contribution: enlarging the farfield from $20c$ to $50c$ *reduces* the computed lift by 2.7%, so the farfield constraint cannot explain the remaining lift deficit relative to experiment at high incidence. The deficit is attributable to the physical modelling chain — fully turbulent boundary layers, a steady closure, and moderate chordwise resolution of the leading-edge suction peak — and is quantified, bounded by the repeats above, and carried through the interpretation of the high-α results.

B Solver convergence histories

Figure 8 shows representative convergence histories. Attached cases meet the residual targets; post-stall cases reach the iteration limit with slowly drifting forces and are tail-averaged as described in Section 3.

C Reproducibility manifest

Every number, figure, and table in this paper is produced by the pipeline in the project repository: `01_fetch_airfoil_coords.py` (geometry), `02_generate_mesh.py` (meshes), `03_prepare_cases.py` (case staging), `04_run_all_cases.sh` (batch execution), `05_postprocess.py` (forces, C_p, C_f, separation), `06_generate_figures.py` (all figures), `07_validate_vs_experiment.py` (validation tables), and `08_build_report.py` (this document). The experimental reference files in `data/experimental/` are the TMR-distributed digitizations of the NACA datasets cited above. Re-running the pipeline end-to-end regenerates every artifact from the scripts alone.

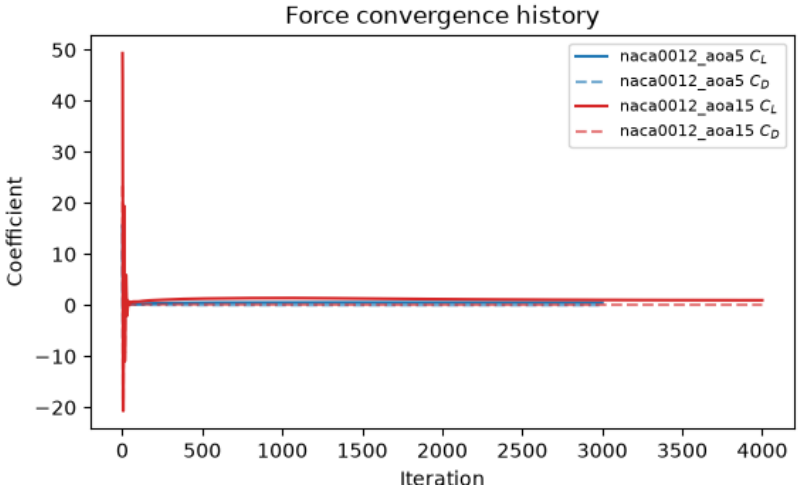


Figure 8: Force-coefficient convergence history, NACA 0012 at $\alpha = 5°$ and $\alpha = 15°$.